



URBAN FUTURES



presented by

**American Museum
of Natural History**



LEAP

in collaboration with



STARR WHITEHOUSE
Landscape Architects
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NVIDIA

FINAL PRESENTATIONS

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New York, NY || January 2026

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East Harlem COAD



Background



East Harlem's
History

Flooding

Historically Harlem has faced flooding problems well before it's conception:

- Harlem Creek connected to a salt marsh that connects to the **Harlem River**
- Harlem Creek would at times flood constructions like the **Lenox Ave** Subway when it was constructed in 1907

Trash

Young Lord's was a young activist group comprised of Puerto Ricans in **East Harlem**:

- **Uncollected trash** lingered for weeks
- Started the **"Garbage Offensive"** initiative in 1969



Problem Define

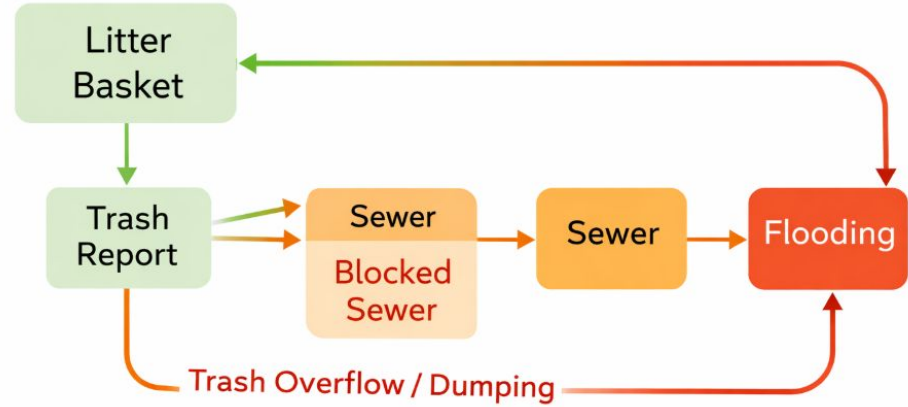


Figure 1. Trash-to-Flood Cascade Model

Small trash overflow, sewer-wide blockage, city-scale flooding.

Data Sources

- 311 service requests 
- NYCDEP catch basins 
- DSNY litter basket locations 
- NYC stormwater flood maps 



How we use it

- Bin density by NTA (GeoPandas)
- Trash hotspot detection (LLM)
- Blockage-risk catch basins
- Overlay with flood-risk blocks (ArcGIS)



ArcGIS

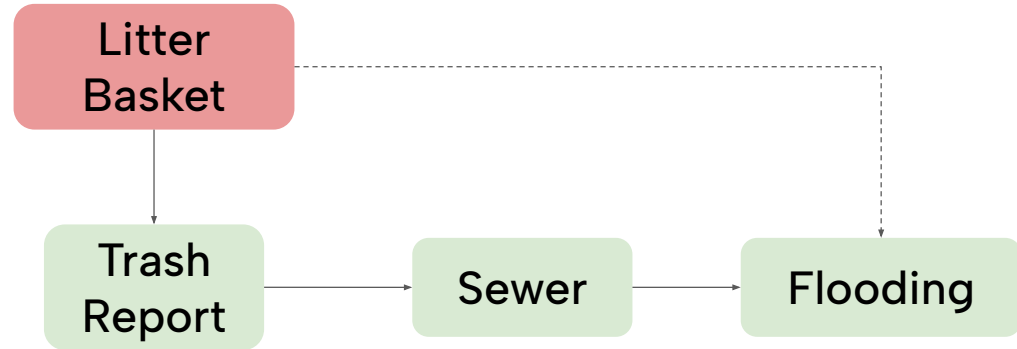
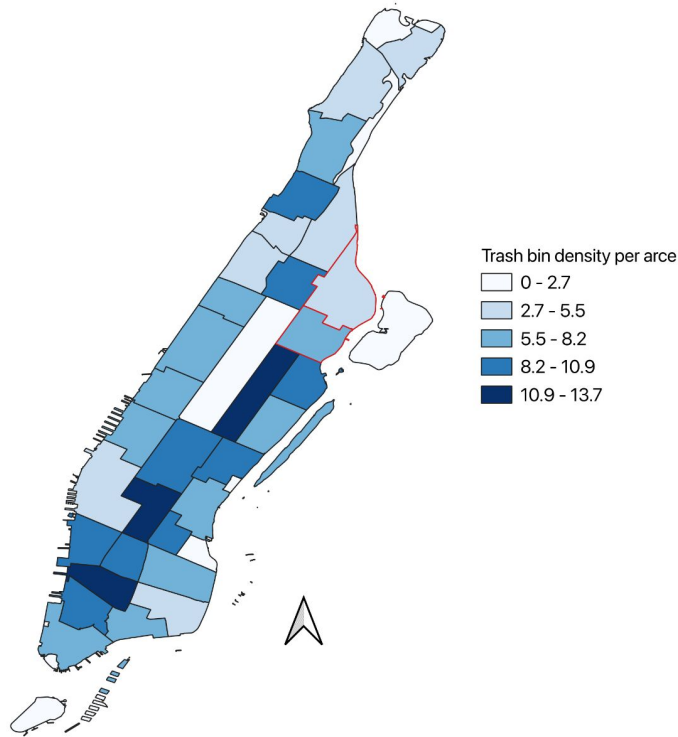


Google AI Studio



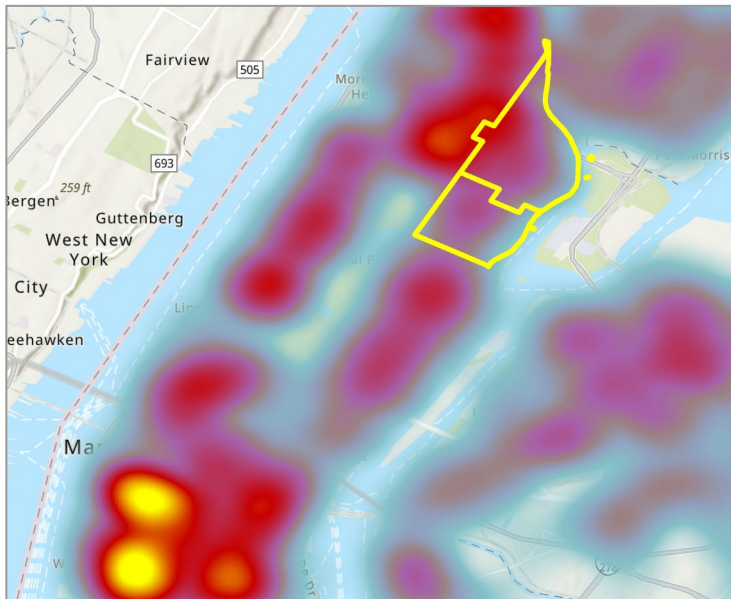
GeoPandas

Results: Quantity + Distribution Gaps of Litter Basket



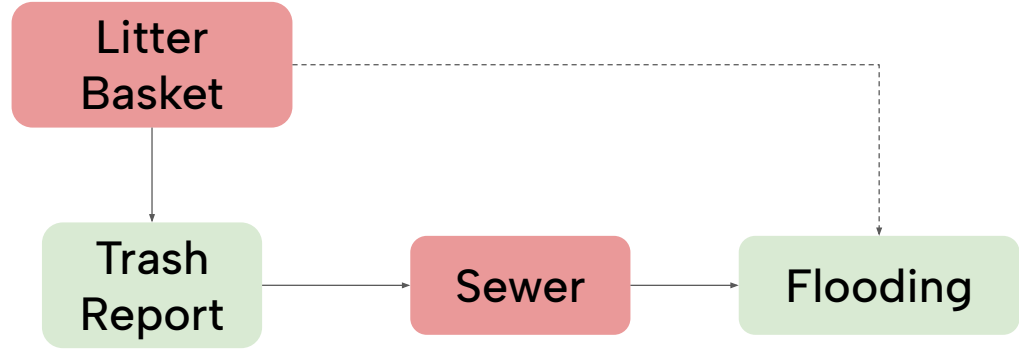
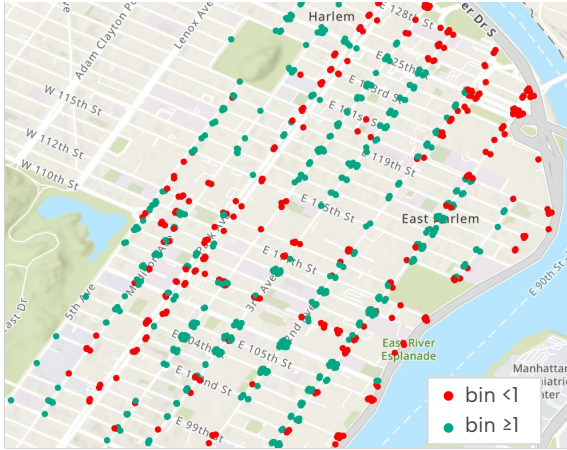
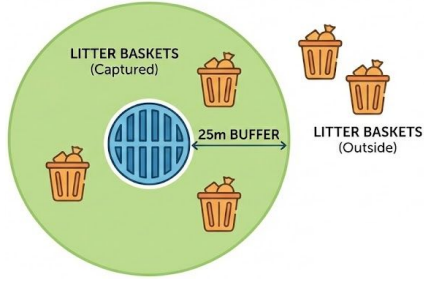
- Average trash bin density in East Harlem is **5.4 bins / acre**
- Average trash bin density in Upper East Side is **9.6 bins / acre**

Results: Harlem's Trash Vulnerability



- Harlem is bearing the brunt of trash related issues while not having the same resources to address such problems
- This costs the city \$\$\$ due to the fact that these areas are very high in traffic and would benefit from clearer streets

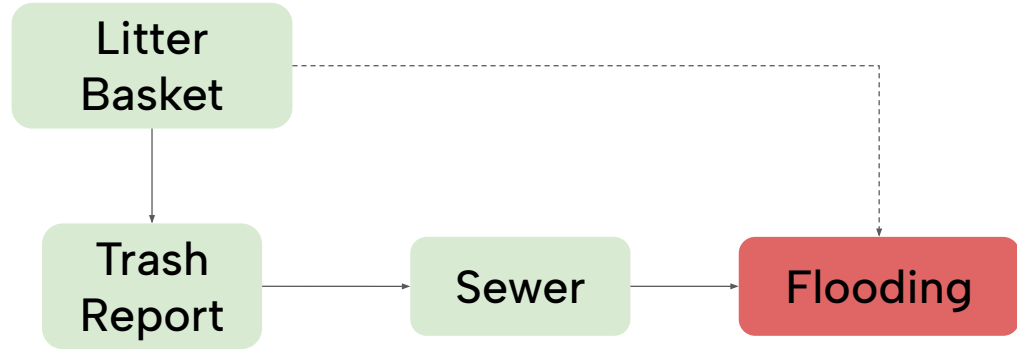
Results: Catch Basin Pandas Identification



- Average distance from catch basin to trash in East Harlem is **46.7 m**
- Average distance from catch basin to trash in Upper East Side is **19.8 m**
- We defined basins as having clogging risk (red dots) if no trash bin falls within a 25 m radius



Results: Clogged Basins Increase Flood Vulnerability



- Flooding map shows potential flooding scenarios under sea level rise conditions
- Stormwater flood area **intersects** with high clogging risk basins
- Future projections show these are **still perceptible**

Interactive Storymap



“A **small** miss today becomes a **BIG** cost tomorrow.”

THANK YOU!

